

OXIDATION REDUCTION TITRATIONS-II

Week 8 Lecture

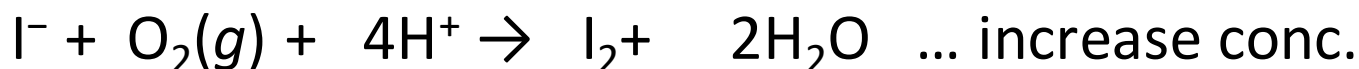
PCC/PCD 105

Standard Iodine solution, I_3^-

- Properties
- weak oxidant... only selective for strong reducing agents
- brown , but not self ind. Except in case of karl-Fischer

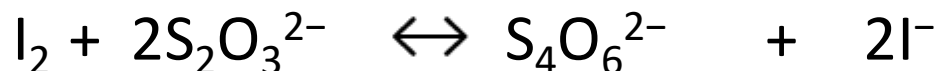


- Stability of standard solution.... 2ry standard



I_2 is volatile ... decrease conc.

- Standardization



Reduction half reaction



- **OR**



- *Eq. wt = M wt/ 2 for 1 mole iodine.*

Titration involving Iodine

Iodimetric titration:

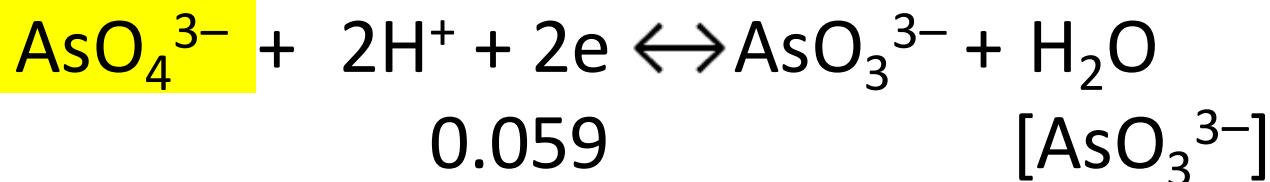
- Direct
- $\text{H}_3\text{AsO}_3 + \text{I}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{AsO}_4 + \text{I}^- + 2\text{H}^+$ (NaHCO₃, starch)
- Ep detection

Starch is used as indicator It is added at the beginning of the titration. The first drop excess of I_3^- (I_2) after the equivalence point causes the solution to turn blue.

N.B. starch is not a true redox indicator

- Conditions:

Titration occurs in neutral or weakly alkaline solutions

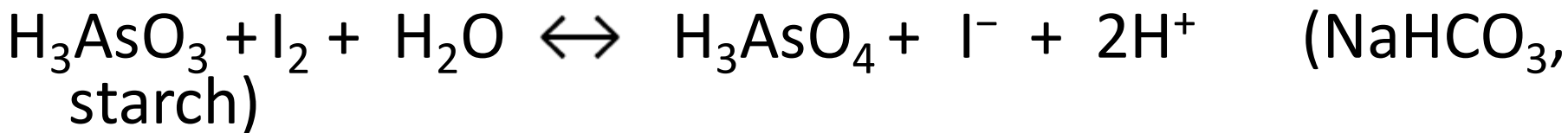


$$E = E^\circ - \frac{0.059}{2} \log \frac{[\text{AsO}_3^{3-}]}{[\text{AsO}_4^{3-}][\text{H}^+]^2}$$

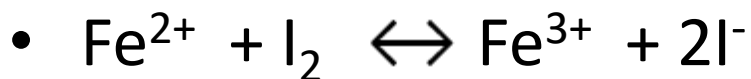
1- $[\text{H}^+] \downarrow$

$E \downarrow$ of the sample

2- Equilibrium shift to the right of the overall reaction.



- **How to force iodine-analyte reaction toward completion**



- **pH effect**



pH has no effect on electrode potential.

- **Complexing agent**

$\text{Fe}^{3+}/\text{Fe}^{2+}$ system,



- Addition of complexing agents as F^- or EDTA that form stable cpx with $\text{Fe}^{3+} \rightarrow$ shift to lower E ($E < 0.53 \text{ V}$) that allows oxidation of Fe^{2+} with I_2

Sources of error in iodimetry

- **Due to I_2**

Lack of stability (volatility, air oxidation of iodide)

- **Due to starch**

Decomposition products,

e.g. -glucose may be oxidized with iodine+(Ve error)

- other products give irreversible reddish color with I_2 .

To Avoid, add preservative like boric acid or formamide.

Applications in Iodimetry

- By Direct Titrations

Determination of $\text{Na}_2\text{S}_2\text{O}_3$



Determination of As^{3+}

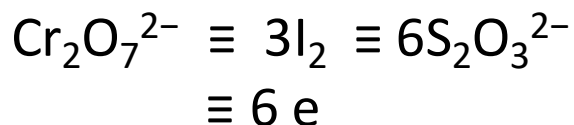
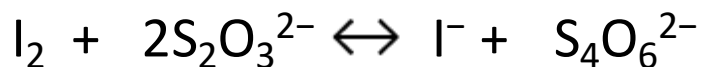
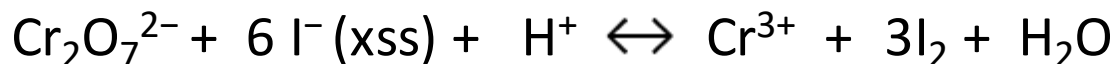


- Starch ind, added at the beginning of titration

- The solution is neutral or slightly alkaline, use NaHCO_3

Iodometric titration:

- Indirect
- For analyzing oxidizing agents
- Add xss I^- to a solution of analyte. I_2 is produced in an amount equivalent to the oxidizing agent. Liberated I_2 is titrated with $Na_2S_2O_3$



$Na_2S_2O_3$ is a universal titrant for I_2 in **neutral or acidic solutions**

- Ep detection

Starch near E.P.... (if added in the beginning will form irreversible blue color)

$CHCl_3$ or CCl_4

Titration occurs in acidic solutions



$$E = E^\circ - \frac{0.059}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-][\text{H}^+]^8}$$

$[\text{H}^+] \uparrow$ $E \uparrow$

(titration should be performed rapidly to minimize air oxidation of iodide)

- How to force oxidation of iodide by analyte toward completion

1-pH

↑ [H⁺] promotes oxidizing agent-iodide reaction (↑ E of oxidizing agent couple)

2- Precipitating agent



$E^0 \text{ Cu}^{2+}/\text{Cu}^{+} = +0.34 \text{ V}$ while $E^0 \text{ I}_2/\text{I}^{-} = +0.536 \text{ V}$

adding SCN⁻ OR I⁻ reagent to ppt Cu⁺ --- [Cu⁺] ↓ →
 ↑ E of Cu²⁺/Cu⁺

$$E = E^0 - \frac{0.059}{1} \log \frac{[\text{Cu}^{+}]}{[\text{Cu}^{2+}]}$$

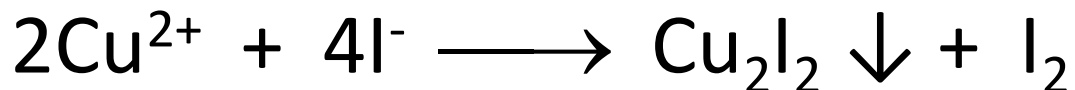
[Cu⁺] ↓

E ↑

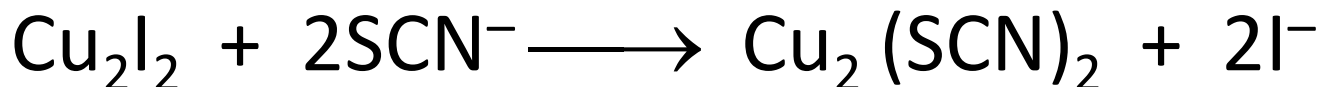
Function of iodide

(a) reduce Cu^{2+} to Cu^+

(b) precipitate Cu^+ ion



But the precipitated Cu_2I_2 would adsorb I_2 on its surface, so we add $(\text{NH}_4)\text{SCN}$ to precipitate the more insoluble $\text{Cu}_2(\text{SCN})_2$ and set free the adsorbed I_2 .

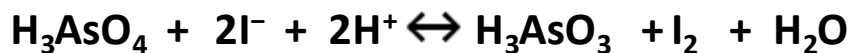
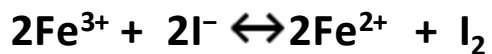
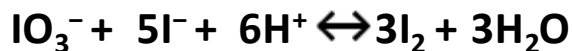
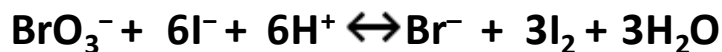
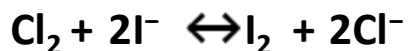
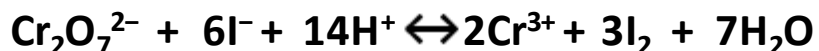
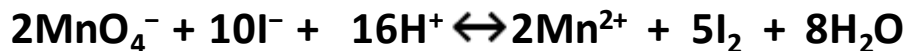


Sources of error in iodometry

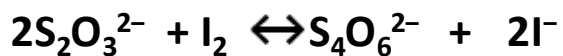
- Decomposition of thiosulfate solution.
- Premature addition of starch and
- Starch decomposition.
- **Due to I₂**
Lack of stability (volatility, air oxidation of iodide)

Iodometric

Determination of oxidizing agents



Titrate the liberated I_2 w/ $\text{Na}_2\text{S}_2\text{O}_3$, CCl_4 ind



Titration involving Iodine

Iodimetry

- Weakly alkaline or neutral solution
- If pH is too alkaline

I₂ will disproportionate to hypoiodide and iodide



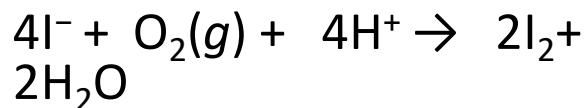
IO⁻ is a stronger oxidant than I₂

Iodometry

- Acidic solution
- Why titration solution kept NOT strongly acidic

Starch ind. tends to hydrolyze in strong acid.

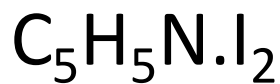
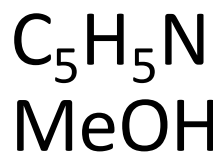
I⁻ tends to be oxidized by dissolved oxygen. In strongly acidic medium



karl Fischer reagent for determination of water

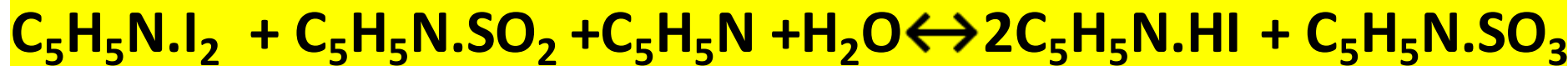
- Reagent

Iodine, sulfur dioxide dissolved in pyridine (I_2 , SO_2 present as complexes *to* prevent their volatilization) and xss. methanol



- Application

Determination of water in organic liquids and solids which are soluble in methanol and can be analyzed by direct titn (acids, alcohols, esters, anhydrides...)



Method: Direct titration

E.p.: potentiometry

Color change from **yellow** (reaction products) to **brown** (xss reagent, pyridine iodine cpx)

Stoichiometry: 1 mole of I_2 , 1 mole of SO_2 , for each mole of H_2O .



Role of methanol in Karl Fisher Reagent

- $\text{C}_5\text{H}_5\text{N}.\text{SO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{C}_5\text{H}_5\text{NH}.\text{SO}_4\text{H}$
- $\text{C}_5\text{H}_5\text{N}.\text{SO}_3 + \text{CH}_3\text{OH} \rightleftharpoons \text{C}_5\text{H}_5\text{NH}.\text{SO}_4\text{CH}_3$
- Prevent reaction of product “sulfur trioxide” with water sample eliminating the susceptible –ve error

